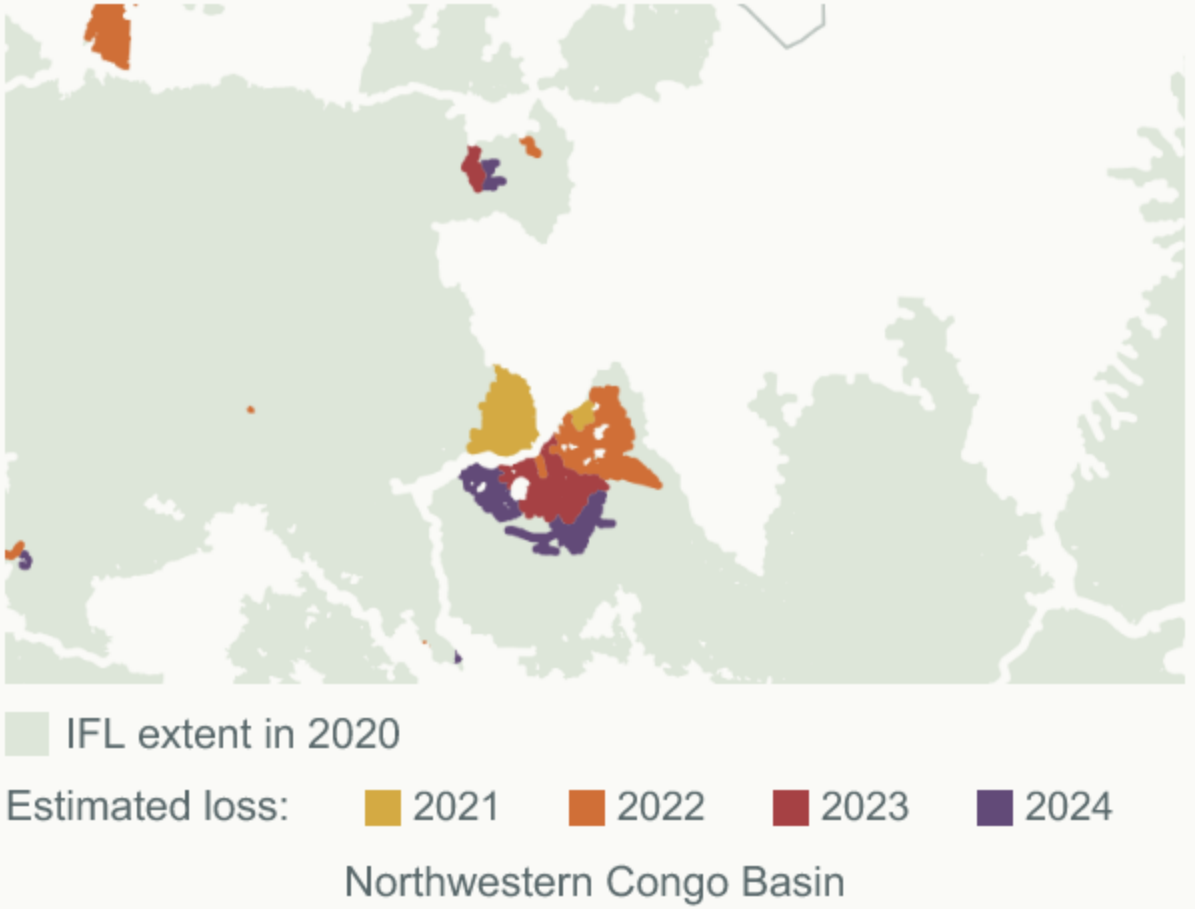
Annual loss of intact forest landscapes

Iddo Glass (2025)

Estimate annual intact forest landscape (IFL) loss between existing map snapshots, 2001-2024.

Developed for Houngbedji et al. (2026) using Google Earth Engine. Meant to examine when IFL loss occurred relative to forest certification.



Houngbedji, K., Isman, J. J., Plakhtieva, M., & Wren-Lewis, L. (2026). What are the Limits to Voluntary Certification? Evidence from an Attempt to Protect Biodiversity. Working paper.

What qualifies as an intact forest landscape?

Large, connected natural ecosystems without substantial human disturbance.

Existing snapshots: 2000, 2013, 2016, 2020 and 2025
(Potapov et al., 2017; IFL Mapping Team, 2025)

2km	500km²	10km
Minimum width of corridors and appendages	Minimum patch area	Minimum width at the broadest point

Potapov et al. (2017). The last frontiers of wilderness: Tracking loss of intact forest landscapes from 2000 to 2013. Science Advances, 3, e1600821.

IFL Mapping Team (2025). Intact Forest Landscapes Extent and Change, 2000-2025.

From disturbance to IFL loss

Detect disturbance → Buffer by 1km

Apply spatial rules

2km

Minimum width of corridors and appendages

Remove narrow parts.

500km²

Minimum patch area

Remove smaller remnants.

10km

Minimum width at the broadest point

Not applied: little added value, high computing cost.

Estimating loss one year at a time



Western Congo, 2021 loss over current satellite imagery

Tree-loss pixels 1km candidate buffers Previous IFL

1. Find tree loss

Select the year's forest loss and exclude mapped fires.

2. Group disturbance

Buffer connected pixels by 1km. Keep groups with enough loss.

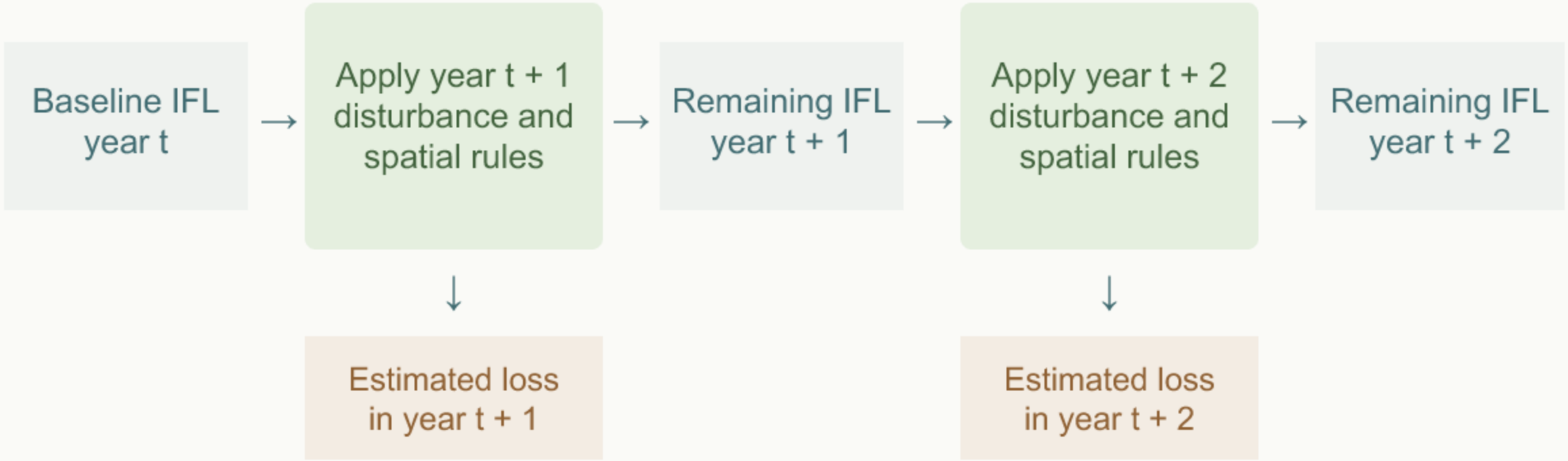
3. Apply IFL rules

Remove narrow parts and small remaining patches.

Hansen et al. (2013). High-resolution global maps of 21st-century forest cover change. Science, 342, 850-853.
Tyukavina et al. (2022). Global trends of forest loss due to fire from 2001 to 2019. Frontiers in Remote Sensing, 3, 825190.

Updating the remaining IFL each year

Each update starts from the area remaining after the previous year.



Implementation at 30m resolution

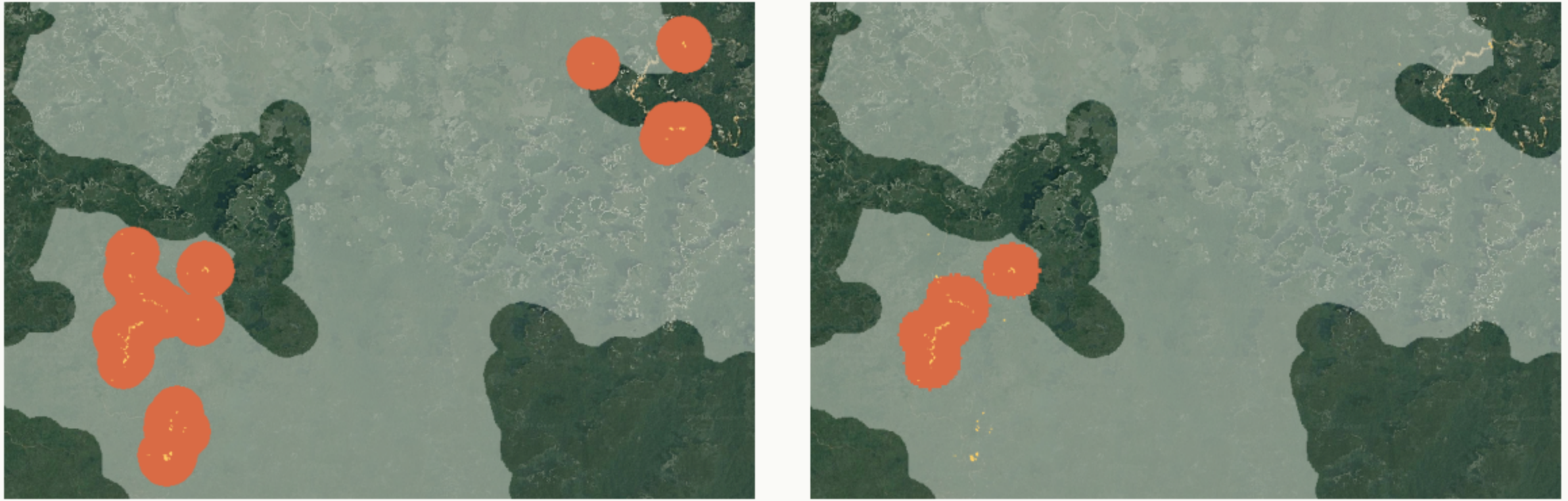
Spatial operations and area calculations at 30m were expensive.
Runs remained costly even after the changes below.

Execution	Developed in the GEE JavaScript editor. Python automated annual runs.
Regions	Split processing into regions, drawing boundaries to avoid cutting IFL patches.
Spatial filtering	Used raster erosion and regrowth.
Area checks	Screened patches at 200m, then removed small areas from the detailed map.

Improving detection: filtering isolated groups

Before: all candidate buffers

After: groups with enough loss pixels



Tree-loss pixels Buffered candidates Previous IFL

Western Congo, 2021 loss over current satellite imagery

Corridor and appendage filtering



Remaining IFL Removed: disturbance Removed: spatial rule

Western Congo, 2021 loss over current satellite imagery

1. Shrink by 1km

Narrow parts disappear.

2. Regrow by 1.5km

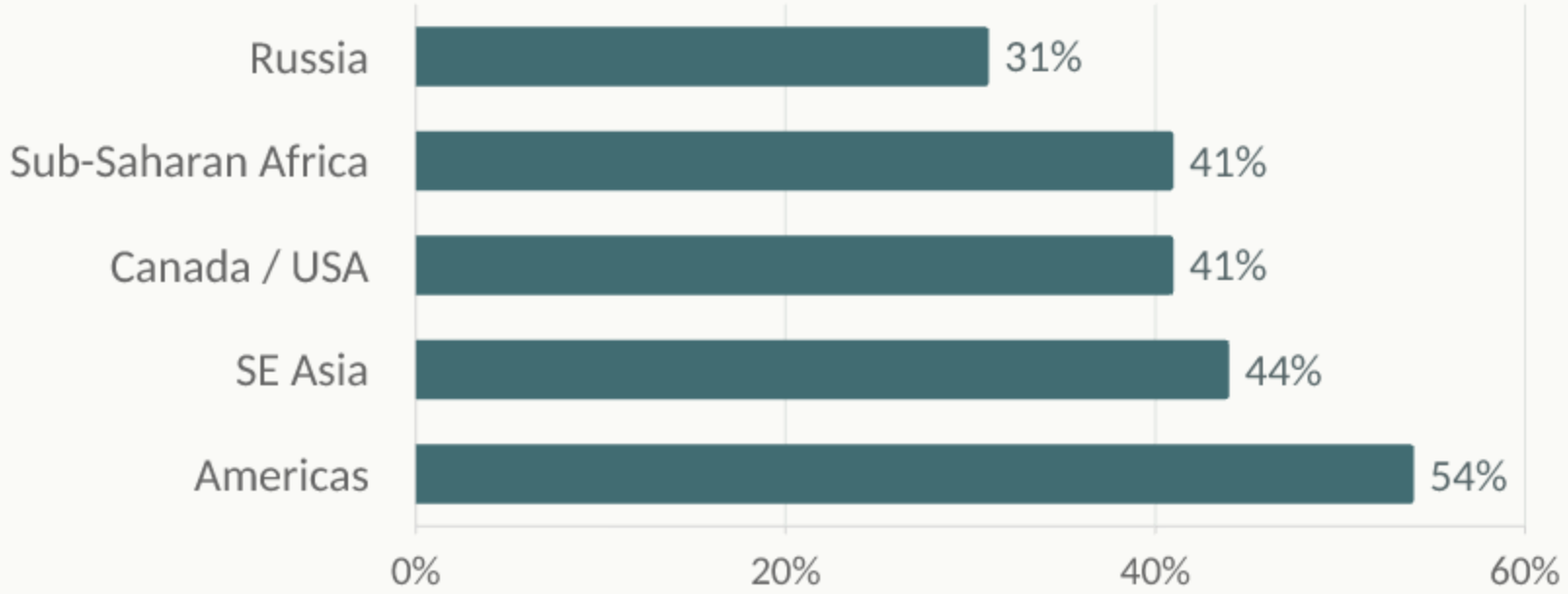
Restore the wider areas.

3. Keep the overlap

Limit regrowth to the original IFL.

Share of snapshot loss assigned to a year

2021-2024 estimates, compared with loss between the 2020 and 2025 IFL maps



Recall

The share of loss shown in the IFL snapshots that the method detects and assigns to a year.

Precision: 98.9-99.6%, nearly 100% by design as estimates are restricted to loss in the IFL snapshots.

Limitations: not just human disturbance

Road



Riverbed



Both count as IFL loss here: both appear as tree-cover loss in Hansen et al. (2013).

Papua New Guinea. Satellite imagery: 2026.

Limitations

- Some loss shown in the IFL snapshots remains unassigned to a year.
- No annual reference maps are available to check the estimated year of loss.
- The fire-loss data ends in 2023, leaving 2024 without this filter.
- Coarser area checks can miss small remnants.
- GEE quotas limited further runs and testing.